

Django Developer Task – Build a Mini ERP System

Objective

Build a simplified ERP module that manages:

- **Products**
 - **Customers**
 - **Sales Orders**
 - **Stock movements**
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ERP Requirements

1 Authentication & Permissions

- Users must log in.
 - Two roles:
 - **Admin** → full access
 - **Sales User** → can only create sales orders, cannot edit stock or products
 - Use Django's built-in auth or DRF token auth.
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2 Product Module

Fields

Field	Type
SKU	CharField (unique)
Name	CharField
Category	CharField
Cost Price	Decimal
Selling Price	Decimal
Stock Qty	Integer

Features

- Admin can create/update/delete products.
 - Sales user can view only.
 - Stock automatically decreases when sales orders are created & paid.
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3 Customer Module

Fields:

- Customer code
- Name
- Phone
- Address
- Email
- Opening balance (optional)

Features:

- CRUD for admin
 - Sales user can add customers but cannot delete
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4 Sales Orders

Sales Order Fields

Field	Type
Order Number	Auto generated
Customer	FK
Order Date	Date
Created By	User
Status	pending / confirmed / cancelled
Total Amount	Calculated

Sales Order Details:

- Product
- Qty
- Price
- Total (qty × price)

Business Logic

- When order is **confirmed**:
 - reduce stock qty from product
 - If order is **cancelled**:
 - stock should return automatically
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5 Stock Movement Log

For every stock change (sales order), log:

- Product
 - Qty
 - User
 - Timestamp
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6 UI / API

Option A → Django Templates

- Simple Bootstrap pages
- Basic CRUD forms
- Table listing products/orders

Option B → Django REST Framework

- REST API endpoints
 - JWT or Token authentication
 - Postman collection or Swagger API docs
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7 Optional Bonus Features

- Export reports to Excel (xlsx)
- Dashboard with:
 - total customers
 - total sales today
 - stock running low
- Role-based menu display
- Product images upload
- Pagination + search + filters